

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - V

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Class : SY IT	Batch :
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Binary Search Tree: Write a program to:

- a. Create a binary search tree.
code

```
class TreeNode:
    def __init__(self, data):
        self.data = data
        self.left = None
        self.right = None

root = TreeNode('R')
nodeA = TreeNode('A')
nodeB = TreeNode('B')
nodeC = TreeNode('C')
nodeD = TreeNode('D')
nodeE = TreeNode('E')
nodeF = TreeNode('F')
nodeG = TreeNode('G')

root.left = nodeA
root.right = nodeB

nodeA.left = nodeC
nodeA.right = nodeD

nodeB.left = nodeE
nodeB.right = nodeF

nodeF.left = nodeG

# Test
print("root.right.left.data:", root.right.left.data)
```

output

```
root.right.left.data: E
```

- b. Insert nodes into a binary search tree.

code

```
def search(root, key):
    if root is None:
        return False
    print("checking:", root.data)

    if root.data == key:
        return True
    return search(root.left, key) or search(root.right, key)

key = input("Enter element to search:").upper()
#Search
if search(root, key):
    print("Element Found")
else:
    print("Element Not Found")
```

output

```
n.py
root.right.left.data: E
Enter element to search:Z
checking: R
checking: A
checking: C
checking: D
checking: B
checking: E
checking: F
checking: G
Element Not Found

root.right.left.data: E
Enter element to search:G
checking: R
checking: A
checking: C
checking: D
checking: B
checking: E
checking: F
checking: G
Element Found
```

c. Search for a node in a binary search tree.

code

```
def search(root, key):
    if root is None:
        return False
    if root.data.upper() == key.upper():
        return True

    return search(root.left, key) or search(root.right, key)

key = input("Enter letter to search: ")

if search(root, key):
    print("Found!")
else:
    print("Not Found!")
```

output

```
root.right.left.data: E
Enter letter to search: A
Found!

= RESTART: C:/Users/IT-52/AppD
n.py
root.right.left.data: E
Enter letter to search: Z
Not Found!
```

Curiosity Questions:

1. In a BST, which side (left or right) has **smaller numbers** than the root?
2. If we insert numbers 10, 20, 5 into a BST, which number will be at the root?
3. What will the inorder traversal of a BST give — sorted numbers or random numbers?
4. Where will the **largest number** in a BST always be found?
5. If the root of a BST is 50 and we search for 70, will we go left or right? Why?
6. If a BST has only **one node**, what is its inorder traversal?
7. Can we create a BST from characters like A, B, C instead of numbers?
8. If we insert numbers 5, 10, 15, 20 in order, will the tree be balanced or like a straight line?
9. In a BST, do we check the left child first or the right child first in inorder traversal?
10. If we delete the root of a BST, will the tree disappear?